System Documentation

Electronics

Electronic Engine Control Unit

ECU 9

Application: FRAC AET with ECU 9, EXU and EMU

Series 4000 T95

E532827/00E

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1 Safety

1.1 Safety regulations for startup and operation

Safety regulations for startup

- Prior to initial operation all official approvals must have been granted and the prerequisites for commissioning met.
- The product must be installed as instructed and accepted to manufacturer's standards before being put into operation for the first time.
- When putting the product into operation, always ensure that:
  - All maintenance and repair work has been completed
  - All loose parts have been removed from rotating machine components
  - All personnel is clear of the danger zone represented by rotating machine components
- Immediately after putting the product into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems are working properly.

Safety regulations for equipment operation

- The operator must be familiar with the controls and displays.
- The operator must be familiar with the consequences of any operations performed.
- During operation, the display instruments and monitoring units must be permanently observed with regard to present operating status, violation of limit values and warning or alarm messages.

Malfunctions and emergency stop

- The procedures for cases of emergency, in particular, emergency stop, must be practiced regularly.
- The following steps must be taken if a malfunction of the system is recognized or reported by the system:
  - Inform supervisor(s) in charge.
  - Analyze the message.
  - If required, carry out emergency operations e.g. emergency stop.

Operation

Note the following requirements before starting the product:
- Wear ear protectors.
- Ensure that the engine room is well-ventilated.
- Do not inhale the exhaust gases of the product.
- The exhaust system is leak-tight and that the gases are vented to atmosphere.
- Mop up any leaked or spilled fluids and lubricants immediately or soak up with a suitable binder agent.
- Protect battery terminals, generator terminals or cables against accidental contact.

Operation of electrical equipment

When electrical equipment is in operation, certain components of these appliances are electrically live.
- Observe the safety instructions for these devices.

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1.2 Safety regulations for maintenance and repair work

Before commencing maintenance and repair work

- Have maintenance or repair work carried out by qualified and authorized personnel only.
- Allow the product to cool down to less than 50°C (risk of explosion of oil vapors, fluids and lubricants, risk of burning).
- Relieve pressure in fluid and lubricant systems and compressed-air lines which are to be opened. Use suitable containers of adequate capacity to catch fluids and lubricants.
- When changing the oil or working on the fuel system, ensure that the engine room is adequately ventilated.
- Never carry out maintenance and repair work with the product in operation.
- Do not test operation with the product running unless expressly instructed to do so.
- Secure the product against unintentional starting, e.g. with start interlock.
- Attach “Do not operate” sign in the operating area or to control equipment.
- Disconnect the battery. Lock contactors.
- Close the main valve on the compressed-air system and vent the compressed-air line when pneumatic starters are fitted.
- Disconnect the control equipment from the product.

In addition for starters equipped with copper beryllium pinions:

- Wear a breathing mask (filter class P2) when performing maintenance work to eliminate the health risk presented by the beryllium in the pinion.
- Do not blow out the interior of the flywheel housing or the starter with compressed air.
- Furthermore, the interior of the flywheel housing must be cleaned with a class H dust extraction device.

During maintenance and repair work

- Take special care when removing ventilation or plug screws from the product.
- Cover the screw or plug with a cloth to prevent fluids escaping under pressure when removing vent screws or plugs.
- Take care when draining hot fluids and lubricants (risk of burning).
- Use only proper and calibrated tools. Observe the specified tightening torques during assembly or disassembly.
- Carry out work only on assemblies or plants which are properly secured.
- Never use lines for climbing.
- Keep fuel injection lines and connections clean.
- Always seal connections with caps or covers if a line is removed or opened.
- Take care not to damage lines, in particular fuel lines, during maintenance and repair work.
- Ensure that all retainers and dampers are installed correctly.
- Ensure that all fuel injection and pressurized oil lines are installed with enough clearance to prevent contact with other components. Do not place fuel or oil lines near hot components.
- Do not touch elastomeric seals if they have carbonized or resinous appearance unless hands are properly protected.
- Note cooling time for components which are heated for installation or removal (risk of burning).
- When working high on the equipment, always use suitable ladders and work platforms.
- Make sure components or assemblies are placed on stable surfaces.
- Ensure particular cleanness during maintenance and repair work on the product. After completion of maintenance and repair work, make sure that no unattached parts are in/on the product (e.g. cloths and cable ties).

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After completing maintenance and repair work

- Ensure that all personnel is clear of the danger zone presented by the product before cranking.
- Check that all guards have been reinstalled and that all tools and loose parts have been removed after working on the product (in particular, the barring tool).

Welding work

Welding on the product or mounted units are not permitted.

- Remove parts (e.g. exhaust pipes) which are to be welded from the product beforehand.
- Cover the product when welding in surrounding areas.

Before starting welding work:
- Switch off the power supply master switch.
- Disconnect the battery.
- Separate the electrical ground of electronic equipment from the ground of the unit.

When welding in the vicinity of the product:
- Do not perform other maintenance and repair work on the product. Risk of explosion or fire due to oil vapors and highly flammable fluids and lubricants.
- Do not use product as ground terminal.
- Never position the welding power supply cable adjacent to, or crossing wiring harnesses of the product. The welding current may otherwise induce an interference voltage in the wiring harnesses which could conceivably damage the electrical system.

Hydraulic installation and removal

- Check satisfactory function and safe operating condition of tools, jigs and fixtures to be used. Use only the specified jigs and fixtures for hydraulic removal/installation procedures.
- Observe the max. permissible force-on pressure specified for the jig/fixture.
- Do not attempt to bend or apply force to lines.
- During hydraulic installation and removal, ensure that nobody is standing in the immediate vicinity of the component to be installed/removed.

Before starting work, pay attention to the following:
- Vent the hydraulic installation/removal tool, the pumps and the lines at the relevant points for the system to be used (e.g. open vent plugs, pump until bubble-free air emerges, close vent plugs).
- For hydraulic installation, screw on the jig with the piston retracted.
- For hydraulic removal, screw on the jig with the piston extended.
- For a hydraulic installation/removal jig with central expansion pressure supply, screw spindle into shaft end until correct sealing is established.

Working with batteries

- Observe the safety instructions of the battery manufacturer when working with batteries.

Gases emanating from the battery are explosive.
- Avoid sparks and naked flames.

- Never place tools on the battery.
- Before connecting the cable to the battery, check the battery polarity.

Battery pole reversal may lead to injury through the sudden discharge of acid or bursting of the battery body.

- Never place tools on the battery.
- Do not allow electrolyte to come in contact with skin or clothing.
- Wear goggles and protective gloves.

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Working on electrical and electronic assemblies

- Always obtain the permission of the person in charge before commencing maintenance and repair work or switching off any part of the electronic system required to do so.
- De-energize the appropriate areas prior to working on assemblies.
- Do not damage cabling during removal work.
- When reinstalling ensure that wiring is not damaged during operation by contact with sharp objects, by rubbing against other components or by a hot surface.
- Do not secure cables on lines carrying fluids.
- Do not use cable straps to secure cables.
- Always use connector pliers to tighten union nuts on connectors.
- Subject the device as well as the product to a function check on completion of all repair work. In particular, check the function of the engine emergency stop feature.
- Store spare parts properly prior to replacement, i.e. protect them against moisture in particular.
- Pack defective electronic components and assemblies in a suitable manner when dispatched for repair, i.e. protected, in particular, against moisture and impact and wrapped in antistatic foil if necessary.

Working with laser equipment

- When working with laser equipment, always wear special laser-protection goggles (hazard due to heavily focused radiation).

Laser equipment must be equipped with the protective devices necessary for safe operation according to type and application.

For conducting light-beam procedures and measurement work, only the following laser devices must be used:
- Laser devices of classes 1, 2 or 3A.
- Laser devices of class 3B, which have maximum output in the visible wavelength range (400 to 700 nm), a maximum output of 5 mW, and in which the beam axis and surface are designed to prevent any risk to the eyes.

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2 Design and Function

2.1 System Layout

2.1.1 Configuration of an electronic engine control system featuring Engine Control Unit ECU 9

Engine Control Unit ECU 9 is the heart of the electronic engine control system. It serves as an interface for engine and plant signals (including parallel-wired signals, e.g. pushbuttons, limit switches and analog sensors, bus-based signals and bus-capable sensors and actuators).

The figure below shows the AET system for Series 4000T95. This system features an ECU 9 to control the engine and an EMU 8 to monitor the individual exhaust gas sensors. Connection to the customer system is established via the MAU interface. A J1939 CAN bus is applied to CAN 2 in this case.

The EXU is used to monitor the characteristics of the injectors and to correct ECU injection timing depending on the condition of the injectors.

Dialog system DiaSys is temporarily connected to CAN 1 at the MAU. It is used to configure Engine Control Unit ECU 9 in the course of initial start-up and later to check and/or modify the existing configuration. Dialog system DiaSys allows extensive modification of Engine Control Unit settings depending on authorization.

NOTICE

Additional information:
Please read the product-specific documentation.
- DiaSys is described in a separate document which accompanies the dialog system.

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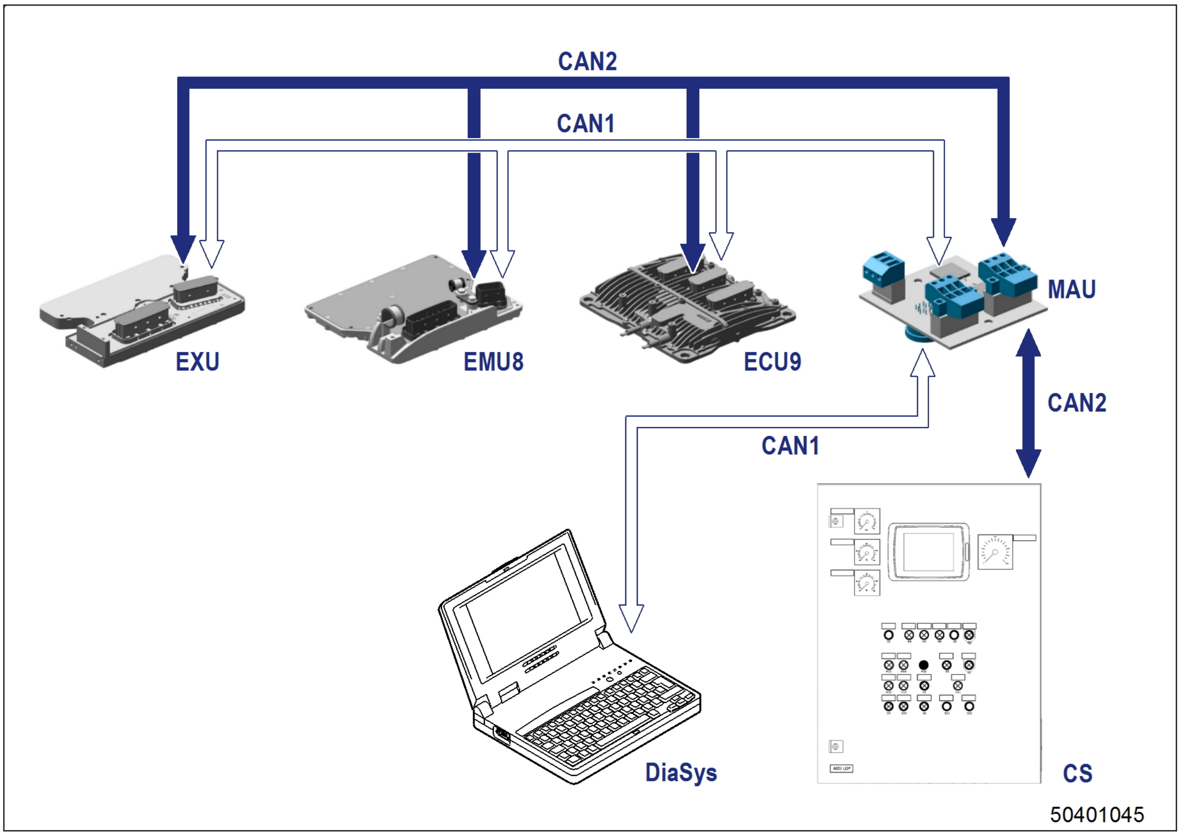

Figure 1: System overview – AET Series 4000T95

ECU 9	Engine Control Unit	CAN2	J1939 CAN bus	EXU	Extension Unit for ECU 9
EMU 8	Engine Monitoring Unit – Individual exhaust gas temperature monitoring	CAN1_E	Engine CAN bus		
		CS	Customer System		
CAN 1	MTU CAN bus with DiaSys connectivity (optional)				

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